Aritra Kundu

Graduate Student | Department of Physics & Astronomy, University of Pennsylvania

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Research Interests

Galaxy formation and evolution, reionization, near-field cosmology, first galaxies, stellar streams, synthetic observations

Education

Ph.D., Physics & Astronomy, University of Pennsylvania — Philadelphia, USA
Advisors: Prof. Robyn Sanderson & Prof. Adam Lidz — 2022 – 2027 (expected)
Thesis: *Probing Reionization using both Low-z and High-z Galaxies in Cosmological Simulations.*

M.Sc., Physics, Presidency University — Kolkata, India
Advisor: Prof. Ritaban Chatterjee — 2019 – 2021
Thesis: *RMS-Flux Relation and Disc-Jet Connection in Blazars in the context of the Internal Shocks Model*

B.Sc., Physics, Presidency University — Kolkata, India
Advisor: Prof. Kanan K. Datta — 2016 – 2019
Thesis: *Cosmic Recombination History in Light of EDGES Measurements of the Cosmic Dawn 21 cm Signal*

List of Publications

Primary Author

(includes 1st and 2nd authored papers)

1. **Kundu**, **A.**, Sanderson, R., Lidz, A., Gandhi, P. J., Wetzel, A., Feldmann, R., et al. (2026), "Rise of the Forsaken Relics: Connecting Present-day Stellar Streams and Phase-mixed Galaxies to the Epoch of Reionization," ApJ, 999, 124.
2. **Kundu**, **A.**, Chatterjee, R., Mitra, K., Mondal, S. (2022), "rms-flux relation and disc-jet connection in blazars in the context of the internal shocks model," MNRAS, 510, 3688.
3. Datta, K. K., **Kundu**, **A.**, Paul, A., Bera, A. (2020), "Cosmic recombination history in light of EDGES measurements of the cosmic dawn 21-cm signal," PRD, 102, 083502.

Co-Authored

1. Mandacarú Guerra, L. S., O'Neil, S., Lisanti, M., Roy, S., Sanderson, R., **Kundu**, **A.**, et al. (Sept. 2026), "Probing Atomic Dark Matter with Stellar Streams in Milky Way-mass Galaxies," ApJ, 1009, 76.
2. Tasso, E., Lucey, M., Sanderson, R., **Kundu**, **A.**, Necib, L. (2026), "The dynamics of old inner galaxy stars in Milky Way-mass galaxies using FIRE-2 simulations," MNRAS, 550, stag1139.
3. Lucey, M., Sanderson, R. E., Horta, D., **Kundu**, **A.**, Hopkins, P. F., Arora, A., et al. (2025), "Cosmological Predictions for Minor Axis Stellar Density Profiles in the Inner Regions of Milky Way–mass Galaxies," ApJ, 982, 87.

Submitted/In Prep Papers

Primary Author

1. **Kundu**, **A.**, Lidz, A., Sun, G. (Sept. 2026), "Probing the Reionization History and Bubble Sizes with JWST Lyman-α Fraction Measurements," arXiv e-prints, arXiv:2609.40363 (*under review in ApJ*).
2. **Kundu**, **A.**, Sanderson, R., Shipp, N., Thob, A., Lidz, A., Panithanpaisal, N. (2026), "Near-Far Constraints on Reionization from Simulated LSST Observations of Dwarf Galaxy Stellar Streams and Satellites (ready for submission),"
3. **Kundu**, **A.**, et al. (2026), "Near-Far Constraints on Reionization from Simulated Roman Observations of Stellar Streams and Satellites in External Galaxies (in prep.),"

Co-Authored

1. Sun, G., Lidz, A., Chang, T.-C., Faucher-Giguère, C.-A., Furlanetto, S. R., **Kundu**, **A.**, et al. (Sept. 2026), "LIMFAST. V. Probing Reionization with the Synergy Between 21 cm and Photometric Galaxy Surveys," arXiv e-prints, arXiv:2609.35957.

Research Presentations

Colloquia & Seminars

1. **Yale Galaxy Lunch**, *'Probing Reionization using Low-z & High-z Galaxies'*, **Nov 2026, Yale University**, New Haven, USA
2. **Galaxy Formation Lunch**, *'Probing Reionization using Low-z & High-z Galaxies'*, **Oct 2026, UC Berkeley**, Berkeley, USA
3. **Galaxy Formation group meeting**, *'Probing Reionization using Low-z & High-z Galaxies'*, **Oct 2026, University of Chicago**, Chicago, USA
4. **CIERA Theory Seminar**, *'Probing Reionization using Low-z & High-z Galaxies'*, **Oct 2026, Northwestern University**, Chicago, USA
5. **Near-field Cosmology group meeting**, *'Probing Reionization using Low-z & High-z Galaxies'*, **Aug 2026, University of Washington**, Seattle, USA
6. **Stellar Streams group meeting**, *'Probing Reionization using Low-z & High-z Galaxies'*, **July 2026, Carnegie Observatories**, Pasadena, USA
7. **UC Davis Physics Special Seminar**, *'Probing Reionization using Low-z & High-z Galaxies'*, **July 2026, UC Davis**, Davis, USA

Conferences & Workshops

1. **Contributed Talk**, *'Roman Lyman-Alpha Emitters at Cosmic Dawn and the Prospects for LIM Cross-Correlations'*, Aug 2026, **Line Intensity Mapping 2026**, California Institute of Technology, Pasadena, USA
2. **Contributed Talk**, *'Rise of the Forsaken Relics: Probing reionization using disrupted galaxies'*, Jul 2026, **Near-field Cosmology in the Era of Big Data: Local Group and Beyond**, Fields Institute, Toronto, Canada

3. **Contributed Talk**, *'Rise of the Forsaken Relics: Probing reionization using disrupted galaxies'*, Jul 2026, **Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman**, California Institute of Technology, Pasadena, USA
4. **Contributed Talk**, *'Rise of the forsaken relics: connecting present-day stellar streams and phase-mixed galaxies to the Epoch of Reionization'*, Jun 2026, **GalFirenze 2026**, Florence, Italy (virtual)
5. **Contributed Poster**, *'Probing reionization with disrupted substructures'*, Jul 2024, **Small Galaxies, Cosmic Questions - II**, Durham University, Durham, UK
6. **Contributed Poster**, *'RMS-Flux relation and disc-jet connection in blazars in the context of the internal shocks model'*, Mar 2022, **40th Meeting of the Astronomical Society of India**, IIT Roorkee & ARIES Nainital, India (virtual)
7. **Contributed Talk**, *'RMS-Flux relation and disc-jet connection in blazars in the context of the internal shocks model'*, Feb 2022, **21st National Space Science Symposium,**, India (virtual)
8. **Contributed Talk**, *'Cosmic recombination history in light of EDGES measurements of the cosmic dawn 21-cm signal'*, Oct 2021, **4th Global 21 cm workshop**, University of Colorado, Boulder, USA (virtual)
9. **Contributed Talk**, *'Cosmic recombination history in light of EDGES measurements of the cosmic dawn 21-cm signal'*, Feb 2021, **39th Meeting of the Astronomical Society of India**, ICTS-TIFR Bengaluru, IISER Mohali, IIT Indore & IUCAA Pune, India (virtual)
10. **Contributed Poster**, *'Impact of colder IGM scenario on the Recombination History of the Universe'*, January 2020, **Observing the First Billion Years of the Universe**, IIT Indore, India

Awards and Fellowships

1. Received the Parthasarathi Gupta Memorial Fund, for being the best student in M.Sc. Physics 2021
2. Gold medalist in M.Sc. Physics 2021
3. Received the Satyabrata Ghosh Cash Prize 2020 in B.Sc. for highest grades in Physics in 2019

Mentoring Experience

Jack Terell, *UPenn undergraduate* 2026 - present
– Mentoring on a project on *Probing reionization and first galaxies with Lyα and Hα measurements from simulations.*

Sara Michele Alvarado Rodriguez, *UPenn undergraduate* 2025 - present
Penn Undergraduate Research Mentoring Program
– Primary mentor on *Cosmological predictions for stellar populations in LSST-detectable tidal streams.*
– Paper to be submitted soon.

Alexandra DiMauro, *UPenn undergraduate* 2024 - present
Penn Undergraduate Research Mentoring Program
– Primary mentor on *Signatures of CDM and SIDM Models in Dwarf Galaxies in FIRE-2 Simulations.*
– Paper to be submitted soon.

Teaching Experience

Guest Lectures

University of Pennsylvania

ASTR 1212, Introduction to Astrophysics II — Spring 2025, 2024
– Lectured on reionization

ASTR 1211, Introduction to Astrophysics I — Fall 2025
– Taught a class on Doppler effect

Teaching Assistantships

University of Pennsylvania

ASTR 1211, Introduction to Astrophysics I — Fall 2026, 2025
– Tutoring Python-lab exercises, grading assignments and exams, holding office hours and exam reviews.

ASTR 3392, Life & Death of Stars — Spring 2026
– Grading assignments and exams.

ASTR 1212, Introduction to Astrophysics II — Spring 2025, 2024, Fall 2022
– Tutoring Python-lab exercises, grading assignments and exams, holding office hours and exam reviews.

ASTR 0007, The Big Bang & Beyond — Fall 2024
– Grading assignments and exams, holding office hours.

ASTR 0001, The Survey of the Universe — Spring 2023
– Grading assignments and exams.

Professional Activities and Outreach

1. **Peer mentoring first-year graduate students** at University of Pennsylvania from 2026-2027, to help them navigate the graduate school.
2. **Organizer of the Departmental Friday Astro Lunch/Journal Club** at the University of Pennsylvania since Fall 2024.
3. **Member of Scientific Organizing Committee**, Presision - Undergraduate Physics Symposium, in 2019 and 2021, Presidency University, Kolkata.
4. **Co-organizer, Presision** - Undergraduate Physics Symposium, in 2017, 2018 and 2020, Presidency University, Kolkata.

Skills

Programming Languages	Python (proficient), Fortran (intermediate), SQL (basic)
Languages	Bengali (native), English (fluent), Hindi (fluent)